SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 3 (Easy): Maximum Profit Analyzer

Problem Description

Given daily profit/loss values, find the maximum possible profit obtainable from a contiguous sequence of days using Kadane's Algorithm.

Input Format

- First line contains integer N.
- Second line contains N integers.

Output Format

Print maximum subarray sum.

Constraints

- $1 \leq N \leq 10^5$

Sample Input

8
-2 -3 4 -1 -2 1 5 -3

Sample Output

7

PROGRAM :

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int maxSum = arr[0];
        int currentSum = arr[0];

        for (int i = 1; i < n; i++) {
            currentSum = Math.max(arr[i], currentSum + arr[i]);
            maxSum = Math.max(maxSum, currentSum);
        }

        System.out.println(maxSum);
    }
}
```

OUTPUT :

```
8
1 2 -5 -6 4 3 4 2
13


...Program finished with exit code 0
Press ENTER to exit console.
```